

The mechanism: one theta identity yields the complete analytic
package,
at every rank and at both archimedean chart types

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Abstract

Rung 9 of the benchmark ladder, stated as compiled theorems rather than a register claim. For every self-dual-multiset Satake bank and every admissible archimedean chart — pure complex $\prod_i \Gamma_{\mathbb{C}}(s + \mu_i)$ or mixed $\Gamma_{\mathbb{R}}(s + \delta) \prod_i \Gamma_{\mathbb{C}}(s + \mu_i)$ — a *single pointwise identity* between the two prescribed theta readouts of the 3D bank machine-yields the complete analytic package: entire continuation, functional equation $\Lambda(1 - s) = \varepsilon \Lambda(s)$, and the chart identification on the initial half-plane. Every other input is compiled: local integrability, rapid decay at both ends, the Mellin identification, the contragredient comparison, and the identity theorem globalization. The mixed case is powered by a new compiled clock: the Gaussian $2x^\delta e^{-\pi x^2}$ with Mellin transform exactly Deligne’s real factor $\Gamma_{\mathbb{R}}(s + \delta)$, convolved in signed-log height with the complex-pair kernel. Instantiating at the symmetric powers of a level-one eigenform seed — the self-duality input is the compiled clock reversal — gives the mechanism at *every rank* Sym^r and, at $r = 4$, finishes rung 5: the Sym^4 package from one identity at its mixed chart $\Gamma_{\mathbb{R}}(s + \delta) \Gamma_{\mathbb{C}}(s + (k - 1)) \Gamma_{\mathbb{C}}(s + 2(k - 1))$, weight $k = 2n$.

1 The mechanism at a complex chart

Fix a finite channel type ι and a polynomial Satake dual pair W on ι : a primal weight system and its contragredient, with radial Euler coefficients $c_j = h_j(\text{weights})$ and $c_j^\vee$ (complete homogeneous symmetric functions), and polynomial growth exponents. The two prescribed 3D readouts are the theta banks of these coefficients against the completion kernel of a shift list μs : the primal bank $\Theta(x) = \sum_n c_n \phi_n(x) g(h_n(x))$ read on the carrier, and the reciprocal-height contragredient bank. These are the objects `cpsPolynomialFullPrimal3DBankReadout` and `cpsPolynomialFullDual3DTransformedReadout` of the corpus; their local integrability, rapid decay, and initial-half-plane Mellin identification with $\prod_i \Gamma_{\mathbb{C}}(s + \mu_i) \cdot D(s)$ are compiled facts, not hypotheses.

Theorem 1 (the mechanism; `ThetaMechanism.mechanism`; machine-checked). *Let W be any polynomial Satake dual pair whose contragredient coefficients equal its primal coefficients ($c_j^\vee =$*

*Machine-checked companions: `RequestProject/GlobalHelixRealClockCompletion.lean` (the real clock and the mixed kernel, 24 named declarations) and `RequestProject/ThetaMechanism.lean` (the mechanism and its instances, 15 named declarations), each declaration at axiom footprint `{propext, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`.

c_j for all j — the self-dual-multiset condition). Let $\mu s \neq \emptyset$ be any shift list with $\operatorname{Re} \mu \geq 0$, and $\varepsilon \neq 0$. Suppose the one identity

$$\Theta\left(\frac{1}{x}\right) = \varepsilon x \cdot \Theta^\vee\left(\frac{1}{x}\right) \quad (x > 0)$$

between the prescribed primal readout and the reciprocal-height contragredient readout. Then the niceness package exists over the shifted bank: an entire Λ with $\Lambda(1-s) = \varepsilon \Lambda(s)$ and $\Lambda(s) = \prod_i \Gamma_{\mathbb{C}}(s + \mu_i) \cdot L(s)$ for $\operatorname{Re} s > B_W$, where B_W is the uniform threshold $\# \iota + \max(\text{exponents}) + 1$ (*ThetaMechanism.mechB*).

The proof consumes nothing beyond the identity: the strong reflection pair assembles with every analytic field compiled; the contragredient transform equals the primal transform by the coefficient equality on the open half-plane and the identity theorem (*entire_eq_of_eqOn_isOpen*); the functional equation and chart follow. The threshold, the summability of the shifted bank beyond it (*ThetaMechanism.bankArithmeticSummable*), and the half-plane conditions on the shifts are all uniform in the data — there is no per-instance analytic input anywhere.

2 The real clock and the mixed kernel

Every completion kernel so far was built from the complex-place clock $2x^\mu e^{-2\pi x}$ with Mellin multiplier $\Gamma_{\mathbb{C}}(s + \mu)$. The even symmetric powers require the real place.

GlobalHelixRealClockCompletion.lean adds it:

Lemma 2 (*gammaRClock_mellin*; machine-checked). *The Gaussian clock $g_{\mathbb{R}}^{(\delta)}(x) = 2x^\delta e^{-\pi x^2}$ has*

$$\int_0^\infty g_{\mathbb{R}}^{(\delta)}(x) x^s \frac{dx}{x} = \pi^{-(s+\delta)/2} \Gamma\left(\frac{s+\delta}{2}\right) = \Gamma_{\mathbb{R}}(s + \delta), \quad \operatorname{Re}(s + \delta) > 0,$$

*exactly Deligne’s real factor. The convergence half-plane is *gammaRClock_mellinConvergent*; the uniform signed-log weighted profile bound is *weightedGammaRLogProfile_bounded*.*

The proof substitutes $u = x^2$ in the carrier coordinate and lands on the Γ -integral; the 2 in the clock absorbs the Jacobian $\frac{1}{2}$. The *mixed kernel* is then one real clock convolved in signed-log height with the complex-pair completion kernel,

$$\text{mixedKernelLog} = g_{\mathbb{R}}^{(\delta)} *_{\mathbb{M}} \star_i 2x^{\mu_i} e^{-2\pi x},$$

and the three kernel facts that power everything are compiled at every δ and every finite list: uniform boundedness at every admissible weight (an L^1 profile convolved with a bounded profile is bounded), decay faster than every power, and the exact multiplier —

weightedMixedKernelLog_bounded, *mixedKernelLog_rapid*, *mixedKernelLog_hasMellin* :

$$\tilde{g}(s) = \Gamma_{\mathbb{R}}(s + \delta) \cdot \prod_i \Gamma_{\mathbb{C}}(s + \mu_i).$$

On this kernel the full 3D bank layer is rebuilt verbatim: the banks *mixedPrimal3DBankReadout* and *mixedDual3DTransformedReadout*, their theta identifications, local integrability, rapid decay, and the two initial-half-plane identifications *mixedPrimal3D_initialIdentification* and *mixedDual3D_initialIdentification*.

3 The mechanism at a mixed chart

Theorem 3 (`ThetaMechanism.mechanismR`; machine-checked). *Let W be self-dual-multiset as in Theorem 1, let δ have $\operatorname{Re} \delta \geq 0$, and let $\mu s \neq \emptyset$ have nonnegative real parts, $\varepsilon \neq 0$. One pointwise identity between the mixed primal readout and the mixed reciprocal-height contragredient readout yields the mixed package (`MixedNicePackage`): entire Λ , $\Lambda(1-s) = \varepsilon \Lambda(s)$, and*

$$\Lambda(s) = \Gamma_{\mathbb{R}}(s + \delta) \cdot \prod_i \Gamma_{\mathbb{C}}(s + \mu_i) \cdot L(s) \quad (\operatorname{Re} s > B_W).$$

The continuation in a mixed package is unique (`MixedNicePackage.unique`).

Theorems 1 and 3 are *one* mechanism at the two archimedean chart types in classical use. Neither knows anything about modular forms: the inputs are a self-dual finite weight multiset, a chart, and one reflection.

4 Every rank of the eigenform seed; the Sym^4 finish

For the Sym^r clock of a level-one weight- $2n$ eigenform seed the self-duality input is not a hypothesis: the contragredient of the Sym^r weight system is its own clock reversal, so $c_j^\vee = c_j$ at *every* rank (`Sym3ThetaWall.symrPair_dual_eq_primal`, consumed here), and the shifted radial coefficients are the Sym^r bank (`symrCoefficientArithmetic_eq`). Instantiating:

Corollary 4 (`symr_package_of_theta`, `symr_mixed_package_of_theta`; machine-checked). *For every rank r , every admissible chart of either type, and every $\varepsilon \neq 0$: one theta identity of the prescribed Sym^r readouts yields the complete package for $L(\operatorname{Sym}^r, s)$ over the compiled Sym^r bank, at the uniform threshold.*

Corollary 5 (`rung 5's finish`; `sym4_package_of_theta`; machine-checked). *Weight $k = 2n$, $n \geq 1$. With the exponent-pair shifts $\mu_1 = k-1$, $\mu_2 = 2(k-1)$ (`sym4Shifts`) and any admissible real shift δ : one pointwise identity of the mixed Sym^4 readouts yields entirety, the functional equation, and*

$$\Lambda(s) = \Gamma_{\mathbb{R}}(s + \delta) \Gamma_{\mathbb{C}}(s + (k-1)) \Gamma_{\mathbb{C}}(s + 2(k-1)) \cdot L(\operatorname{Sym}^4, s)$$

on the initial half-plane.

The complex shifts are pure representation theory — the nonzero exponent pairs $\pm(k-1)$, $\pm 2(k-1)$ of Sym^4 of the discrete-series parameter — and match the classical completions (for $k = 12$: $\Gamma_{\mathbb{C}}(s+11) \Gamma_{\mathbb{C}}(s+22)$, the $\operatorname{Sym}^4 \Delta$ chart; compare Shimura's Sym^2 chart $\Gamma_{\mathbb{R}}(s+1) \Gamma_{\mathbb{C}}(s+k-1)$ [2], whose complex shift is the $i = 1$ case of the same law $\mu_i = i(k-1)$). The real shift δ is carried parametrically: the middle exponent of Sym^{2m} is 0 and contributes $\Gamma_{\mathbb{R}}(s + \delta)$ with $\delta \in \{0, 1\}$ fixed by the archimedean parity of the middle line; the theorem consumes whichever (δ, ε) satisfies the identity — the identity itself pins the chart, no table is consulted. This mirrors the corpus's own register at the identification proposition: one $\Gamma_{\mathbb{R}}(s + \delta)$ adjoined for even r (`universal.tex`, Theorem `prop:localid` — the global identification of the candidate with $\operatorname{Sym}^r \pi$).

With this, the ladder state is: rung 4 (Sym^3) finished from one identity at its pure-complex chart (previous note); rung 5 (Sym^4) finished from one identity at its mixed chart (Corollary 5); and rung 9 — *a new mechanism* — stated and proven as the two generic theorems above, of which every per-rank finish is a two-line instance.

5 Register

Remark 6 (register). ***Proved (machine-checked).** Everything named above: the real clock and its Mellin identification; the mixed kernel’s bounds, decay, and multiplier; the mixed 3D bank layer; the two mechanism theorems; uniqueness; the all-rank and Sym^4 instances. All at axiom footprint $\{\text{proptext}, \text{Classical.choice}, \text{Quot.sound}\}$, no sorry.*

***Consumed from the corpus.** The compiled theta/Mellin engine (theta banks of polynomial coefficients, identification, decay); the compiled clock-reversal self-duality of Sym^r ; the compiled seed data of the eigenform (*HeckeEigenData*).*

***Cited, not formalized.** The classical discharge of the one remaining identity per even rung — automorphy of Sym^4 (Kim [3], Kim–Shahidi [4]) implies the reflection classically; Sym^2 is Gelbart–Jacquet [1] with Shimura’s chart [2]. These are named for scope, not consumed.*

***Not claimed.** The truth of the theta identity for any $r \geq 2$ is not claimed here; it is the isolated input, exactly one pointwise equation per rung. The value of δ for a given even rank is not asserted; the theorem is uniform in it.*

***Falsifier.** Any self-dual-multiset bank with a chart of nonnegative shifts satisfying the one identity whose L -readout fails entirety, the functional equation, or the chart identification on the stated half-plane would refute the mechanism; so would a mixed kernel whose Mellin multiplier deviates from $\Gamma_{\mathbb{R}}(s + \delta) \prod_i \Gamma_{\mathbb{C}}(s + \mu_i)$ at any admissible point.*

References

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